Qi Wei (Wilson) Chen

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Data Science student building machine learning systems and research software for biology, from multicamera 3D pose estimation of animal behavior to whole slide image analysis for pan-cancer pathology.

Education: University of Toronto, B.Sc., Data Science Specialist, Computer Science Major, GPA: 3.8 ***June 2027***

Research Experience:

Cognitive Neuroscience Researcher, *HHMI Janelia Research Campus* ***Summers 2025 & 2026***

Beehavior: Studying Learning and Collaborative Behavior in Bumblebees. PIs: Dr. Yoshi Aso, Dr. Rob Johnson

- Designed and built a six camera, 300 fps 3D behavior rig for bumblebees, with a PTP timestamp resync method that aligns all six video streams onto a single timeline
- Developed staged, glass aware camera calibration through the plexiglass arena, validated to 15 µm against a machined target; reduced reprojection error from 1.42 to 0.90 px and shape error from 0.18% to 0.08%
- Built an LSF cluster harness that ran JARVIS 3D pose inference with TensorRT on 165 recordings totaling ~40 TB of video, using a rolling storage buffer and an automated frame desync gate
- Built a galvo tracking rig in which three YOLO cameras triangulate a walking bee and two pan tilt mirrors steer a 135 mm telephoto camera onto it in real time
- Designed custom 3D CAD arenas for observational learning and other cognitive tasks; developed custom syringe pump firmware from scratch for µL reward dosing

Student Researcher, *Kumar Lab, PMCRT* ***August 2025 — Present***

Pan-cancer computational pathology and tumor evolution in Genomics England. PI: Dr. Sushant Kumar

- Building a containerized, multimodel whole slide image pipeline that extracts cell composition and tissue features from H&E slides, run on AWS through Genomics England's Lifebit platform
- Benchmarking tissue segmentation and nucleus classification models such as HistoPLUS and HoVer-NeXt on runtime and slide level agreement, including a blinded pathologist review of model predictions
- Automated ETL of BRCA tumor sequencing data and built lineage tree reconstruction of clonal evolution

Publications & Software:

- Ispizua JI, …, **Chen W**, …, Johnson RE (17 authors). Whole-body 3D kinematics of freely behaving *Drosophila*. *bioRxiv*, 2026.
  Markerless 50 keypoint 3D pose from seven 800 fps cameras, refined with biomechanical inverse kinematics.
- Yan J, Deb D, **Chen W**, et al. *orange*, v2.1.0. Zenodo, 2026.
  GPU accelerated capture of synchronized multicamera video with real time YOLO detection.
- Yan J, Deb D, **Chen W**, et al. *red*, v1.1.0. Zenodo, 2026.
  Annotation tool for labeling multicamera footage recorded with orange.

Work Experience:

ML External Project Lead, *UTMIST, Lawbrokr* ***August 2025 — January 2026***

- Led 7 developers building Justice AI, an AI assistant SaaS that prescreens clients for law firms
- Built ETL pipelines and RAG based LLMs over internal and external data, deployed with Docker

Teaching Assistant, *Mathematics Department, University of Toronto* ***September 2025 — Present***

- Lead weekly calculus tutorials for 30+ students; grade with Quercus, Crowdmark, and Gradescope

Projects:

CRISPRight: Optimizing sgRNA for CRISPR with Genetic Algorithm *(TensorFlow, Python)* ***September 2025***

- Engineered a predictive pipeline for CRISPR/Cas9 off-target effects using ~230k sequences from 74 studies, integrating DNABERT embeddings, MLP, and a genetic algorithm to evolve optimized sgRNAs

Technical Skills: Python, C++, R, SQL, Java; PyTorch, TensorRT, OpenCV, YOLO; LSF/HPC clusters, Docker, AWS, Git, Arduino; Computer Vision, 3D Pose Estimation, Deep Learning, ETL Pipelines